

Siddhant Sharma

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Research Interests

Optimal control and path planning for autonomous aerial vehicles | Flight dynamics and control | Physics-informed machine learning for aviation safety

Education

Indian Institute of Technology Bombay Mumbai, India
B.Tech, Aerospace Engineering (Minor in AI/ML, ongoing) 2025 – 2029
• CGPA: 9.21/10

Maharaja Sawai Man Singh Vidyalaya, Jaipur Jaipur, India
Class XII (CBSE) — 93% | Class X (CBSE) — 94.2% 2019 – 2025

Research Experience

Optimal Control of Dubins Vehicles Aug 2026 – Present
Undergraduate Researcher, MAV Lab, IIT Bombay | Advisor: Prof. Deepak Prakash Kumar Manuscript in preparation
• Building on the lower-bounding framework for the two-point **Dubins interval problem** by Satyanarayana G. Manyam et al. (J. Intell. Robot. Syst., 2017), which assumes normal extremals.
• **Initial contribution:** extended the analysis to **abnormal extremals** using **Pontryagin's Minimum Principle**; derived the abnormal-case candidate paths and showed that the eight forced-abnormal π -arc paths are non-optimal.
• Implementing the analysis in Python; developing further contributions, with planned **hardware validation on a fixed-wing VTOL UAV** being acquired by the lab.

Engineering Projects

4-DOF Aircraft Flight Dynamics Simulator | *Python, SciPy, NumPy* Jan 2026 – Present
• Nonlinear 4-DOF longitudinal simulator for a Boeing 737, integrating the equations of motion with RK45 using aerodynamic and propulsion models.
• Implemented ISA atmosphere, aerodynamic-coefficient interpolation and compressibility corrections; validated with unit tests against flight-dynamics principles, as a base for autopilot design.

Non-collocated Control of a Torsion Disc System | *AE 308 Course Project, Python* Autumn 2026
• Modelling and controller design for angular positioning of the non-collocated disc of an ECP torsional plant, in a team of four.

650 mm Quadcopter Build | *Tarot Iron Man TL65B01 frame* 2026
• Assembling a 650 mm folding quadcopter platform from a frame kit.

Software Experience

Developer, DevCom IIT Bombay Apr 2025 – Present
React, TypeScript, Python, MySQL, Docker
• **ReaCH:** full-stack modules for IIT Bombay's Research and Career Hub, serving 10,000+ students and recruiters.
• **Resume Builder:** LLM-based parsing of user text into validated JSON, rendered to LaTeX via a containerised pipeline producing PDFs in under 3s.

Linux Telemetry Daemon | *Rust, SQLite* May 2026 – Jun 2026
• Rust daemon collecting system and per-process statistics from `/proc`, with SQLite persistence and CI testing.

Skills & Coursework

Methods: Optimal control (Pontryagin's Minimum Principle), flight dynamics & performance, classical control, numerical integration
Programming: Python (NumPy, SciPy, Matplotlib, Pandas), C++, Rust (introductory), JavaScript/TypeScript
Tools: Git, Linux, Docker, Autodesk Fusion 360, Arduino, OpenCV
Coursework: Control Theory (AE 308), Data Science (DS 203), Computer Programming (CS 101)

Achievements

JEE Advanced 2025 — All India Rank **2279** | **JEE Main 2025** — All India Rank **1182** (99.929 percentile)
Winner, Games of Codes by Replit — Techfest IIT Bombay (2022)
Finalist, AtomQuest by Atomberg — Techfest IIT Bombay (2023)